Here is a step-by-step mathematical derivation of Variant 1 (single bounce with asymmetric leakage), focusing first on the primordial black hole (PBH) lifecycle from formation up to the phase shift, and then deriving the existence of negative primordial black holes (negPBH) from the phase shift onward.

PBH Lifecycle Up to Phase Shift

Assume a single bounce at t=0, where quantum fluctuations seed density perturbations δρ/ρ ≈ 10^{-5} in the hot plasma (T > 10^{12} K). PBH form when these perturbations collapse if their size exceeds the Jeans length $\lambda_J = c_s / \sqrt{G \rho}$, with c_s the sound speed ≈ c/√3 in radiation domination.

The collapse condition is met for fluctuations with mass $M > M_{min} = (4\pi/3) \rho \lambda_J^3$, leading to PBH number density $n_{PBH} = (\delta\rho/\rho)^2 \times (\rho / M_{min})$, or fraction $f_{PBH} = n_{PBH} M_{min} / \rho$.

PBH lifetime before evaporation is governed by Hawking radiation rate $dM/dt = - (\hbar c^4 / G^2 M^2)$, integrated as:

$$t_{evap} = \int_{M_{init}}^{0} (G^2 M^2 / \hbar c^4) \, dM \approx (G^2 / 3 \hbar c^4) M_{init}^3$$

For $M_{init} \approx 10^{15}$ g (typical for bounce-seeded PBH), $t_{evap} \approx 10^{-3}$ s, matching the hot-phase duration $t_{hot} \approx 10^{-3}$ s.

Up to the phase shift at $t_{shift} = t_{hot}$, PBH evaporate partially, losing fraction $\Delta M/M_{init} \approx 1 - \exp(-t / t_{evap})$. The surviving mass at t_{shift} is:

$$M_{surviv} \approx M_{init} \exp(-t_{shift} / t_{evap})$$

with most small PBH fully evaporating ($M_{surviv} \to 0$ for $M_{init} < (3 \hbar c^4 t_{shift} / G^2)^{1/3}$), leaving only larger ones (seeded in anti-matter rich regions) to persist.

Derivation of negPBH Existence from Phase Shift

At t_{shift}, the cooling induces a chiral phase shift $\phi_{shift} = \int H(t) \, dt$ over t_{shift}, where $H(t)$ is the expansion rate $H = \sqrt{8\pi G \rho / 3}$.

This shift winds the geometry, creating a topological asymmetry: $\phi_{shift} = 2\pi N_{wind} + \delta\phi$, with N_{wind} integer windings and $\delta\phi$ the residual phase.

For $\delta\phi > \pi/2$ (strong shift), the winding entangles surviving PBH horizons with the inverted-signature sector, transforming them into negPBH trapped in the higher-dimensional halo.

The transformation probability $P_{neg} = \sin^2(\phi_{shift} / 2) \approx 1 - \exp(- (\delta\phi / \pi)^2)$, approaching 1 for large $\delta\phi$.

The negPBH mass inherits from survivors but inverts signature:

$$M_{neg} = M_{surviv} \times \mathrm{sign}(-g_{(-)})$$, where $g_{(-)} < 0$ is the negative metric.

Existence follows from conservation: the phase shift must balance energy across sectors, requiring negPBH to absorb the anti-matter excess $\Delta M_{anti} = f_{anti} \times \rho \, t_{shift}$, with $f_{anti} \approx 0.5$ in symmetric bounce.

Thus, $N_{negPBH} = \Delta M_{anti} / M_{neg}$, ensuring no local PBH remnants (evaporated or transformed) while sourcing halo effects via leakage.

Post-shift, negPBH function as stable traps, converting trapped mass at rate $dM_{neg}/dt = - \beta M_{neg} / t_{leak}$, with $\beta = f_{mirror}$ (asymmetry factor → 1), balancing universes by funneling energy to the positive side without backflow.